

Piyush Mudgal

New Delhi, India · +91 8178283764 · pranyasharma55555@gmail.com · GitHub: Pyasma · LinkedIn

EDUCATION

Dronacharya College of Engineering, Gurugram

Nov 2021 – July 2025

Bachelor of Technology in Computer Science and Engineering (AI/ML) · Gurugram, Haryana

Coursework: Data Structures & Algorithms, Distributed Systems, Neural Networks, Operating Systems · CGPA: 7.2

INTERNSHIP / EXPERIENCE

ML Intern (Robotics & Computer Vision) | Kyushu Institute of Technology | Fukuoka, Japan Feb 2025 – May 2025

- Built a vision-based environment perception system for robotic arms using Python and Intel RealSense, achieving 95% object detection accuracy with real-time distance estimation.
- Developed a CNN + LSTM gesture recognition pipeline trained on a 100K+ hand gesture dataset, enabling real-time robotic control with 90%+ accuracy.
- Reduced model training time by 60% by optimizing data preprocessing and feature engineering pipelines — co-authored a research paper accepted at ICIEV/IVPR (oral + poster presentation).

Software Engineer (Product & Developer Experience) | Complye (YC-backed) | Remote Nov 2025 – Feb 2026

- Built the Complye Apps Showcase from scratch — a full-stack platform for discovering and sharing community-built applications — enabling self-serve publishing for the first time.
- Reduced developer onboarding friction by authoring end-to-end technical guides covering project setup, deployment workflows, repository structure, and Git practices, adopted as the canonical reference for all new contributors.
- Developed core frontend features for browsing, submitting, and showcasing applications. (TypeScript, REST APIs)

Open Source Contributor | Apache Airflow (Apache Software Foundation) | Global 2025 – Present

- Shipped 10+ merged PRs across providers, testing infrastructure, and core — including a multi-worker race condition fix via atomic file locking, SQLAlchemy URL property implementations for multiple provider hooks, and Playwright E2E test stability improvements using locator-based waits and web-first assertions. (Python, SQLAlchemy, Pytest, Playwright)
- Built a GSoC 2026 proof-of-concept (PR #63876) enabling LLM agents to autonomously complete 7/7 Airflow contribution scenarios with zero manual intervention — by embedding RST agent-skill directives into contributing docs as a single source of truth with a CI drift-detection pre-commit hook. (Python, RST, CI/CD, Claude Code)
- Collaborated with core maintainers Jarek Potiuk and Jason Liu on architecture trade-offs, shortening review cycles by iterating on design decisions via Slack and GitHub before opening PRs.

PROJECTS

Robot Hand Environment Planning | Python, OpenAI Vision API, Intel RealSense, DBSCAN Clustering

- Designed an environment-aware robotic grasping system using Intel RealSense depth sensing and OpenAI Vision API; applied DBSCAN clustering for keypoint selection, achieving successful task completion in dynamic, unstructured environments.

Employee Management System (AI-based) | Java, Spring Boot, PostgreSQL, Docker, AWS, Google Gemini (LLM)

- Built a full-stack system managing 100+ employee records with Gemini LLM natural-language queries, PostgreSQL persistent storage, Docker containerization, and AWS deployment — reducing ad-hoc lookup time versus raw SQL queries.

SKILLS

Programming: Java, Python, R, SQL, HTML, CSS, TypeScript, Data Structures, Algorithms

Backend / DevOps: Spring Boot, REST APIs, Spring Data JPA, JDBC, Microservices, Docker, AWS, CI/CD, pre-commit hooks

AI / Machine Learning: PyTorch, Scikit-Learn, Neural Networks, CNN, RNN, LSTM, RAG, LLMs, Prompt Engineering, Gemini, OpenAI API, Vector Databases

Databases & Tools: PostgreSQL, MongoDB, Git, GitHub, Postman, SQLAlchemy, Playwright

Linux & Open Source: Arch Linux (daily driver), GNOME, AUR, Apache Airflow (Breeze dev environment)

ACHIEVEMENTS

- Research paper accepted at ICIEV/IVPR conference (oral/poster presentation) — contributing to advancements in robotics and computer vision, co-authored during international ML internship at Kyushu Institute of Technology, Japan.
- GSoC 2026 applicant, Apache Airflow — Agent Skill Framework proposal under active mentorship by core maintainers; PoC PR live with 7/7 DX test scenario results measured via Claude Code.
- Selected for international ML research internship in Japan as an undergraduate — sole student from institution placed at Kyushu Institute of Technology robotics lab.